

WI3730TU – Numerical Analysis for PDE's

Lecture 10

Nonlinear Problems:

Picard's method, Banach Fixed-Point Theorem
Newton-Raphson algorithm

Nonlinear Problems

Simple nonlinear (quasi-linear) PDE's

$$\frac{\partial u}{\partial t} = f(u, t)$$

Some famous *diffusion-reaction* equations:

- Fisher's equation (spreading of biological populations):

$$\frac{\partial u}{\partial t} = \Delta u + u(1 - u)$$

- Newell-Whitehead-Segel equation (Rayleigh-Bénard convection):

$$\frac{\partial u}{\partial t} = \Delta u + u(1 - u^2)$$

- Zeldovich equation (combustion theory):

$$\frac{\partial u}{\partial t} = \Delta u + u(1 - u)(u - \alpha)$$

Semi-discrete form of PDE's (systems of ODE's)

After FDM/FVM spatial discretization, nonlinear equations have the form:

$$\mathbf{u}' = \mathbf{f}(\mathbf{u}, t)$$

where $\mathbf{f}(\mathbf{u}, t) \in \mathbb{R}^n$ is a nonlinear vector function of $\mathbf{u} \in \mathbb{R}^n$.

- Fisher's equation:

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} \circ (1 - \mathbf{u})$$

- Newell-Whitehead-Segel equation:

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} \circ (1 - \mathbf{u} \circ \mathbf{u})$$

- Zeldovich equation:

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} \circ (1 - \mathbf{u}) \circ (\mathbf{u} - \alpha)$$

Notation

Here, A denotes the (positive) Laplacian matrix and $\mathbf{u} \circ \mathbf{v}$ denotes the Hadamard (pointwise) product of two vectors:

$$\mathbf{u} \circ \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \circ \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 \\ u_2 v_2 \\ \vdots \\ u_n v_n \end{bmatrix}$$

So, that, for example, $\mathbf{u}^3 = \mathbf{u} \circ \mathbf{u} \circ \mathbf{u}$. Scalar addition:

$$\mathbf{u} + \alpha = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} \alpha \\ \alpha \\ \vdots \\ \alpha \end{bmatrix} = \begin{bmatrix} u_1 + \alpha \\ u_2 + \alpha \\ \vdots \\ u_n + \alpha \end{bmatrix}$$

Two-component reaction-diffusion systems

FitzHugh-Nagumo equation (activator-inhibitor system), two coupled nonlinear PDE's:

$$\begin{aligned}\frac{\partial u}{\partial t} &= D_u \Delta u + \lambda u - u^3 - \kappa - \sigma v \\ \tau \frac{\partial v}{\partial t} &= D_v \Delta v + u - v\end{aligned}$$

After discretization in space, two coupled systems of nonlinear ODE's:

$$\begin{aligned}\mathbf{u}' &= -D_u A \mathbf{u} + \lambda \mathbf{u} - \mathbf{u}^3 - \kappa - \sigma \mathbf{v} \\ \tau \mathbf{v}' &= -D_v A \mathbf{v} + \mathbf{u} - \mathbf{v}\end{aligned}$$

Time-integration: Forward-Euler

The Forward-Euler time-integration method can be applied to any nonlinear system of ODE's of the form $\mathbf{u}' = \mathbf{f}(\mathbf{u}, t)$. FE iterations are:

$$\begin{aligned}\mathbf{u}(t_{k+1}) &= \mathbf{u}(t_k) + h_t \mathbf{f}(t_k, \mathbf{u}(t_k)), \quad k = 0, 1, \dots \\ \mathbf{u}^{k+1} &= \mathbf{u}^k + h_t \mathbf{f}^k(\mathbf{u}^k), \quad k = 0, 1, \dots\end{aligned}$$

For FE it does not matter if $\mathbf{f}^k(\mathbf{u}^k)$ is a *nonlinear* function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, but we know that FE may become unstable and requires sufficiently small time step h_t .

Forward-Euler algorithm for Fisher's equation

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2, \quad \mathbf{u}(t_0) = \mathbf{u}^0$$

Recognize that $\mathbf{f}(\mathbf{u}) = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2$.

Given $\mathbf{u}^0$, predict $\mathbf{u}^{k+1}$ at t_k , $k = 0, 1, \dots, K$, with $h_t = t_{k+1} - t_k$.

$k = 0$

while $k < K$ do :

$$\mathbf{u}^{k+1} = h_t \left[-A\mathbf{u}^k + \mathbf{u}^k - (\mathbf{u}^k)^2 \right] + \mathbf{u}^k$$

$k = k + 1$

Time-integration: Backward-Euler

Consider the Backward-Euler method applied to $\mathbf{u}' = \mathbf{f}(\mathbf{u}, t)$:

$$\mathbf{u}(t_{k+1}) = \mathbf{u}(t_k) + h_t \mathbf{f}(t_{k+1}, \mathbf{u}(t_{k+1})), \quad k = 0, 1, \dots$$

$$\mathbf{u}^{k+1} = \mathbf{u}^k + h_t \mathbf{f}^{k+1}(\mathbf{u}^{k+1}), \quad k = 0, 1, \dots$$

If $\mathbf{f}^k(\mathbf{u}^k)$ is a *nonlinear* function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, then to find $\mathbf{u}^{k+1}$ we need to solve the nonlinear problem

$$\mathbf{u}^{k+1} = h_t \mathbf{f}^{k+1}(\mathbf{u}^{k+1}) + \mathbf{u}^k$$

at each time step k . Here $\mathbf{u}^k$ is a known vector and $\mathbf{f}^{k+1}(\dots)$ is a known function.

Backward-Euler method for Fisher's equation

After spatial discretization we have:

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2$$

Hence, $\mathbf{f}(\mathbf{u}) = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2$ and the BE iterations are

$$\mathbf{u}^{k+1} = \mathbf{u}^k + h_t \left[-A\mathbf{u}^{k+1} + \mathbf{u}^{k+1} - (\mathbf{u}^{k+1})^2 \right], \quad k = 0, 1, \dots$$

Must be solved for $\mathbf{u}^{k+1}$ at each iteration.

Picard's method

Iterative solution of nonlinear problems

Any nonlinear problem can be re-written as

$$\mathbf{u} = h \mathbf{f}(\mathbf{u}) + \mathbf{v}$$

We want to construct an *iterative* method:

$\mathbf{u}_0$ some given initial guess

$$\mathbf{u}_{i+1} = \mathbf{p}(\mathbf{u}_i), \quad i = 0, 1, \dots$$

with the following properties:

- *Convergence*: $\lim_{i \rightarrow \infty} \mathbf{u}_i = \tilde{\mathbf{u}}$, where $\tilde{\mathbf{u}} = h \mathbf{f}(\tilde{\mathbf{u}}) + \mathbf{v}$
- *Small cost*: computing $\mathbf{p}(\mathbf{u}_i)$ is computationally not expensive
- *Fast rate of convergence*: progression to the limit is rapid

Picard's Method

The most simple choice of $\mathbf{p}(\mathbf{u})$ is

$$\mathbf{p}(\mathbf{u}) = h \mathbf{f}(\mathbf{u}) + \mathbf{v}$$

Picard's Method (functional iteration, fixed-point iteration):

- ① choose initial guess $\mathbf{u}_0$
- ② for $i = 0, 1, 2, \dots$ do until convergence:

$$\mathbf{u}_{i+1} = h \mathbf{f}(\mathbf{u}_i) + \mathbf{v}$$

- ③ $\mathbf{u} = \mathbf{u}_{i+1}$

Time-stepping iterations (over k) are called *outer iterations*

Iterations of the nonlinear solver (over i) are called *inner iterations*

Measure of convergence: $\|\mathbf{v} + h \mathbf{f}(\mathbf{u}_{i+1}) - \mathbf{u}_{i+1}\|$ – error in the solution of the nonlinear equation.

Backward-Euler-Picard's algorithm for Fisher's equation

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2, \quad \mathbf{u}(t_0) = \mathbf{u}^0$$

$$\mathbf{u}^{k+1} = \mathbf{u}^k + h_t \left[-A\mathbf{u}^{k+1} + \mathbf{u}^{k+1} - (\mathbf{u}^{k+1})^2 \right], \quad k = 0, 1, \dots$$

Solving for $\mathbf{u}^k$ at t_k , $k = 0, 1, \dots, K$, with $h_t = t_{k+1} - t_k$.

$k = 0$

while $k < K$ do :

$$\mathbf{u}_0^{k+1} = \mathbf{u}^k$$

$$i = 0$$

while $\|\mathbf{u}^k + h_t [-A\mathbf{u}_i^{k+1} + \mathbf{u}_i^{k+1} - (\mathbf{u}_i^{k+1})^2] - \mathbf{u}_i^{k+1}\| > \epsilon$ do :

$$\mathbf{u}_{i+1}^{k+1} = h_t [-A\mathbf{u}_i^{k+1} + \mathbf{u}_i^{k+1} - (\mathbf{u}_i^{k+1})^2] + \mathbf{u}^k$$

$$i = i + 1$$

$$\mathbf{u}^{k+1} = \mathbf{u}_{i+1}^{k+1}$$

$$k = k + 1$$

Banach Fixed-Point Theorem

Theorem: Let $h > 0$, $\mathbf{u}_0 \in \mathbb{R}^n$, and suppose that there exist numbers $\lambda \in (0, 1)$ and $\rho > 0$ such that

- (i) $\|\mathbf{f}(\mathbf{w}) - \mathbf{f}(\mathbf{u})\| \leq \frac{\lambda}{h} \|\mathbf{w} - \mathbf{u}\|$ for every $\mathbf{w}, \mathbf{u} \in \mathcal{B}_\rho(\mathbf{u}_0)$, where $\|\cdot\|$ is a vector norm, and $\mathcal{B}_\rho(\mathbf{u}_0) := \{\mathbf{u} \in \mathbb{R}^n \mid \|\mathbf{u} - \mathbf{u}_0\| \leq \rho\}$ is the closed ball of radius ρ centered at $\mathbf{u}_0$
- (ii) $\mathbf{u}_1 \in \mathcal{B}_{(1-\lambda)\rho}(\mathbf{u}_0)$

Then, for the iterations $\mathbf{u}_{i+1} = h\mathbf{f}(\mathbf{u}_i) + \mathbf{v}$, $i = 0, 1, \dots$ the following holds:

- (a) $\mathbf{u}_i \in \mathcal{B}_\rho(\mathbf{u}_0)$ for every $i = 0, 1, \dots$
- (b) $\tilde{\mathbf{u}} : \lim_{i \rightarrow \infty} \mathbf{u}_i$ exists, satisfies equation $\tilde{\mathbf{u}} = h\mathbf{f}(\tilde{\mathbf{u}}) + \mathbf{v}$, and $\tilde{\mathbf{u}} \in \mathcal{B}_\rho(\mathbf{u}_0)$
- (c) no other point in $\mathcal{B}_\rho(\mathbf{u}_0)$ is a solution of $\mathbf{u} = h\mathbf{f}(\mathbf{u}) + \mathbf{v}$

Proof (1)

Proof: We use induction to show that

$$\|\mathbf{u}_{i+1} - \mathbf{u}_i\| \leq \lambda^i(1 - \lambda)\rho$$

and that $\mathbf{u}_{i+1} \in \mathcal{B}_\rho(\mathbf{u}_0)$ for all $i = 0, 1, \dots$. It is true for $i = 0$ due to (ii), i.e., $\|\mathbf{u}_1 - \mathbf{u}_0\| \leq (1 - \lambda)\rho$. Assume it is true for all $m = 0, 1, \dots, i - 1$, i.e., $\|\mathbf{u}_i - \mathbf{u}_{i-1}\| \leq \lambda^{i-1}(1 - \lambda)\rho$. Then, due to (i):

$$\begin{aligned}\|\mathbf{u}_{i+1} - \mathbf{u}_i\| &= \|[h\mathbf{f}(\mathbf{u}_i) + \mathbf{v}] - [h\mathbf{f}(\mathbf{u}_{i-1}) + \mathbf{v}]\| = h\|\mathbf{f}(\mathbf{u}_i) - \mathbf{f}(\mathbf{u}_{i-1})\| \\ &\leq \lambda\|\mathbf{u}_i - \mathbf{u}_{i-1}\| \leq \lambda\lambda^{i-1}(1 - \lambda)\rho = \lambda^i(1 - \lambda)\rho\end{aligned}$$

This carries the induction from $i - 1$ to i . Consider the sum:

$$\mathbf{u}_{i+1} - \mathbf{u}_0 = \mathbf{u}_{i+1} - \mathbf{u}_i + \mathbf{u}_i - \mathbf{u}_{i-1} + \dots + \mathbf{u}_1 - \mathbf{u}_0 = \sum_{j=0}^i (\mathbf{u}_{j+1} - \mathbf{u}_j)$$

By triangle inequality for a vector norm

$$\|\mathbf{u}_{i+1} - \mathbf{u}_0\| = \left\| \sum_{j=0}^i (\mathbf{u}_{j+1} - \mathbf{u}_j) \right\| \leq \sum_{j=0}^i \|\mathbf{u}_{j+1} - \mathbf{u}_j\|, \quad i = 0, 1, \dots$$

Proof (2)

Exploiting $\|\mathbf{u}_{i+1} - \mathbf{u}_i\| \leq \lambda^i(1 - \lambda)\rho$ and the sum of a geometric series $\sum_{j=0}^i \lambda^j = (1 - \lambda^{i+1})/(1 - \lambda)$,

$$\|\mathbf{u}_{i+1} - \mathbf{u}_0\| \leq \sum_{j=0}^i \lambda^j(1 - \lambda)\rho = (1 - \lambda^{i+1})\rho \leq \rho, \quad i = 0, 1, \dots$$

Hence, $\mathbf{u}_{i+1} \in \mathcal{B}_\rho(\mathbf{u}_0)$ and (a) is true.

Now consider:

$$\begin{aligned} \|\mathbf{u}_{i+k} - \mathbf{u}_i\| &= \left\| \sum_{j=0}^{k-1} (\mathbf{u}_{i+j+1} - \mathbf{u}_{i+j}) \right\| \leq \sum_{j=0}^{k-1} \|\mathbf{u}_{i+j+1} - \mathbf{u}_{i+j}\| \\ &\leq \sum_{j=0}^{k-1} \lambda^{i+j}(1 - \lambda)\rho = \lambda^i(1 - \lambda^k)\rho, \quad i = 0, 1, \dots; \quad k = 1, 2, \dots \end{aligned}$$

Proof (3)

Since $\lambda \in (0, 1)$ and $\|\mathbf{u}_{i+k} - \mathbf{u}_i\| \leq \lambda^i(1 - \lambda^k)\rho$, for any $\epsilon > 0$ we may choose N large enough so that that

$$\|\mathbf{u}_{i+k} - \mathbf{u}_i\| < \epsilon$$

for all $i, (i+k) > N$. In other words, $\{\mathbf{u}_i\}$ is a Cauchy sequence. Since the set $\mathcal{B}_\rho(\mathbf{u}_0)$ is compact (closed and bounded), the Cauchy sequence $\{\mathbf{u}_i\}$ converges to a limit within this set. This proves the existence of $\lim_{i \rightarrow \infty} \mathbf{u}_i = \tilde{\mathbf{u}} \in \mathcal{B}_\rho(\mathbf{u}_0)$.

This limit is a solution of the equation $\mathbf{u} = h\mathbf{f}(\mathbf{u}) + \mathbf{v}$, since:

$$\tilde{\mathbf{u}} = \lim_{i \rightarrow \infty} \mathbf{u}_{i+1} = \lim_{i \rightarrow \infty} [h\mathbf{f}(\mathbf{u}_i) + \mathbf{v}] = h\mathbf{f}(\lim_{i \rightarrow \infty} \mathbf{u}_i) + \mathbf{v} = h\mathbf{f}(\tilde{\mathbf{u}}) + \mathbf{v}.$$

Assume that there exists another $\mathbf{u}^* \in \mathcal{B}_\rho(\mathbf{u}_0)$, $\mathbf{u}^* \neq \tilde{\mathbf{u}}$, such that $\mathbf{u}^* = h\mathbf{f}(\mathbf{u}^*) + \mathbf{v}$. In which case $\|\mathbf{u}^* - \tilde{\mathbf{u}}\| > 0$. On the other hand,

$$\begin{aligned}\|\mathbf{u}^* - \tilde{\mathbf{u}}\| &= \|[h\mathbf{f}(\mathbf{u}^*) + \mathbf{v}] - [h\mathbf{f}(\tilde{\mathbf{u}}) + \mathbf{v}]\| = h\|\mathbf{f}(\mathbf{u}^*) - \mathbf{f}(\tilde{\mathbf{u}})\| \\ &\leq \lambda\|\mathbf{u}^* - \tilde{\mathbf{u}}\| < \|\mathbf{u}^* - \tilde{\mathbf{u}}\|, \quad \text{i.e.} \quad \|\mathbf{u}^* - \tilde{\mathbf{u}}\| < \|\mathbf{u}^* - \tilde{\mathbf{u}}\|\end{aligned}$$

which is not possible. Hence, $\|\mathbf{u}^* - \tilde{\mathbf{u}}\| = 0$ and $\tilde{\mathbf{u}}$ is unique in $\mathcal{B}_\rho(\mathbf{u}_0)$. □

Inner vs outer iterations

- In the outer iterations of the Backward Euler Method:

$$\mathbf{u}^{k+1} = h_t \mathbf{f}^{k+1}(\mathbf{u}^{k+1}) + \mathbf{u}^k$$

it is tempting to take a large h_t

- In the inner iterations of the Picard Method:

$$\mathbf{u}_{i+1}^{k+1} = h_t \mathbf{f}^{k+1}(\mathbf{u}_i^{k+1}) + \mathbf{u}^k$$

it is often required to have a small $h = h_t$. Why? Suppose, $\|\mathbf{f}(\mathbf{u}) - \mathbf{f}(\mathbf{w})\| \leq \xi \|\mathbf{u} - \mathbf{w}\|$ for all $\mathbf{u}, \mathbf{w} \in \mathcal{B}_\rho(\mathbf{u}_0)$, where $\xi > 1$. We can always choose $h_t > 0$ small enough that $\xi = \lambda/h_t$ with $\lambda \in (0, 1)$

- Approach: start with some reasonable h_t . If inner iterations converge very slowly (more than 10-20 iterations), disregard this h_t and restart inner iterations with a smaller h_t
- It is important to have a good initial guess $\mathbf{u}_0^{k+1}$, e.g. $\mathbf{u}_0^{k+1} = \mathbf{u}^k$

Newton-Raphson algorithm

Idea behind the Newton-Raphson method

We want to solve

$$\mathbf{u} = h\mathbf{f}(\mathbf{u}) + \mathbf{v}$$

We have some initial guess $\mathbf{u}_i$. Ideally, to get from $\mathbf{u}_i$ to $\mathbf{u}$, we need to know the difference $\mathbf{u} - \mathbf{u}_i$, since trivially

$$\mathbf{u} = \mathbf{u}_i + \mathbf{u} - \mathbf{u}_i$$

Newton-Raphson's algorithm finds an approximation for $\mathbf{u} - \mathbf{u}_i$

$$\mathbf{u} - \mathbf{u}_i \approx \mathbf{p}_i(\mathbf{u}_i)$$

and iterates as follows:

$$\mathbf{u}_{i+1} = \mathbf{u}_i + \mathbf{p}_i(\mathbf{u}_i)$$

Compare this to a fixed-point iteration (Picard's method): $\mathbf{u}_{i+1} = \mathbf{p}(\mathbf{u}_i)$.

Derivation via Taylor's expansion

$$\mathbf{u} = h\mathbf{f}(\mathbf{u}) + \mathbf{v},$$

$$\mathbf{f}(\mathbf{u}) = \mathbf{f}(\mathbf{u}_i) + \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i} (\mathbf{u} - \mathbf{u}_i) + \mathcal{O}(\|\mathbf{u} - \mathbf{u}_i\|^2),$$

$$\mathbf{u} = h \left[\mathbf{f}(\mathbf{u}_i) + \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i} (\mathbf{u} - \mathbf{u}_i) + \mathcal{O}(\|\mathbf{u} - \mathbf{u}_i\|^2) \right] + \mathbf{v}$$

$$\mathbf{u} - \mathbf{u}_i = h\mathbf{f}(\mathbf{u}_i) + h \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i} (\mathbf{u} - \mathbf{u}_i) + \mathcal{O}(h\|\mathbf{u} - \mathbf{u}_i\|^2) + \mathbf{v} - \mathbf{u}_i$$

$$\mathbf{u} - \mathbf{u}_i - h \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i} (\mathbf{u} - \mathbf{u}_i) = h\mathbf{f}(\mathbf{u}_i) + \mathbf{v} - \mathbf{u}_i + \mathcal{O}(h\|\mathbf{u} - \mathbf{u}_i\|^2)$$

$$\left[I - h \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i} \right] (\mathbf{u} - \mathbf{u}_i) \approx \mathbf{v} + h\mathbf{f}(\mathbf{u}_i) - \mathbf{u}_i$$

$$\mathbf{u} - \mathbf{u}_i \approx \left[I - h \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i} \right]^{-1} [\mathbf{v} + h\mathbf{f}(\mathbf{u}_i) - \mathbf{u}_i]$$

Jacobian

Newton-Raphson's iterations are

$$\mathbf{u}_{i+1} = \mathbf{u}_i + \mathbf{p}_i(\mathbf{u}_i)$$

with

$$\mathbf{p}_i(\mathbf{u}_i) = [\mathbf{I} - hJ(\mathbf{u}_i)]^{-1} [\mathbf{v} + h\mathbf{f}(\mathbf{u}_i) - \mathbf{u}_i]$$

If $\mathbf{u} \in \mathbb{R}^n$ and $\mathbf{f}(\mathbf{u}) : \mathbb{R}^n \rightarrow \mathbb{R}^n$, then the Jacobian matrix $J \in \mathbb{R}^{n \times n}$ is defined as

$$J(\mathbf{u}_i) = \left. \frac{\partial \mathbf{f}(\mathbf{u})}{\partial \mathbf{u}} \right|_{\mathbf{u}=\mathbf{u}_i}$$
$$[J(\mathbf{u}_i)]_{qr} = \left. \frac{\partial f_q(\mathbf{u})}{\partial u_r} \right|_{\mathbf{u}=\mathbf{u}_i}$$

Convergence and modifications

- As long as $h > 0$ is sufficiently small, Newton-Raphson's algorithm has quadratic convergence, i.e., there exists a constant $c > 0$, such that, for sufficiently large i ,

$$\|\tilde{\mathbf{u}} - \mathbf{u}_{i+1}\| \leq c \|\tilde{\mathbf{u}} - \mathbf{u}_i\|^2$$

- With $J = 0$, we get Picard's algorithm:

$$\mathbf{u}_{i+1} = \mathbf{u}_i + [I - h0]^{-1} [\mathbf{v} + h\mathbf{f}(\mathbf{u}_i) - \mathbf{u}_i] = h\mathbf{f}(\mathbf{u}_i) + \mathbf{v}$$

- With $J(\mathbf{u}_i) \approx J(\mathbf{u}_0)$ we get Modified Newton-Raphson's algorithm:

$$\mathbf{u}_{i+1} = \mathbf{u}_i + [I - hJ(\mathbf{u}_0)]^{-1} [\mathbf{v} + h\mathbf{f}(\mathbf{u}_i) - \mathbf{u}_i]$$

Backward-Euler-Newton-Raphson's algorithm for Fisher's equation

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2, \quad \mathbf{u}(t_0) = \mathbf{u}^0$$

Solving for $\mathbf{u}^k$ at t_k , $k = 0, 1, \dots, K$, with $h_t = t_{k+1} - t_k$;
 $\mathbf{f}(\mathbf{u}) = -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2$, and $\mathbf{v} = \mathbf{u}^k$

$k = 0$

while $k < K$ do :

$$\mathbf{u}_0^{k+1} = \mathbf{u}^k$$

$i = 0$

while $\|\mathbf{u}^k + h_t \mathbf{f}(\mathbf{u}_i^{k+1}) - \mathbf{u}_i^{k+1}\| > \epsilon$ do :

$$\mathbf{u}_{i+1}^{k+1} = \mathbf{u}_i^{k+1} + [I - h_t J(\mathbf{u}_i^{k+1})]^{-1} [\mathbf{u}^k + h_t \mathbf{f}(\mathbf{u}_i^{k+1}) - \mathbf{u}_i^{k+1}]$$

$i = i + 1$

$$\mathbf{u}^{k+1} = \mathbf{u}_{i+1}^{k+1}$$

$k = k + 1$

Jacobian matrix of the Fisher's equation

$$\begin{aligned}\mathbf{f}(\mathbf{u}) &= -A\mathbf{u} + \mathbf{u} - \mathbf{u}^2 \\ [J(\mathbf{u})]_{q,r} &= \frac{\partial f_q(\mathbf{u})}{\partial u_r} = \frac{\partial}{\partial u_r} \left[-\sum_{j=1}^n a_{q,j} u_j + u_q - u_q^2 \right] \\ &= -\sum_{j=1}^n a_{q,j} \frac{\partial u_j}{\partial u_r} + \frac{\partial u_q}{\partial u_r} - 2u_q \frac{\partial u_q}{\partial u_r} \\ &= -\sum_{j=1}^n a_{q,j} \delta_{j,r} + \delta_{qr} - 2u_q \delta_{qr} \\ &= -a_{qr} + \delta_{qr} - 2u_q \delta_{qr} \\ J(\mathbf{u}_i^{k+1}) &= -A + I - 2 \operatorname{diag}(\mathbf{u}_i^{k+1})\end{aligned}$$

where $\delta_{q,r}$ is the Kronecker delta function, $I \in \mathbb{R}^{n \times n}$ is the identity matrix, and $\operatorname{diag}(\mathbf{u}_i^{k+1})$ is the matrix with the vector $\mathbf{u}_i^{k+1}$ on its diagonal.

Inner iterations require a linear solver

To compute the update vector

$$\mathbf{p}_i(\mathbf{u}_i^{k+1}) = \left[I - h_t J(\mathbf{u}_i^{k+1}) \right]^{-1} \left[\mathbf{u}^k + h_t \mathbf{f}(\mathbf{u}_i^{k+1}) - \mathbf{u}_i^{k+1} \right]$$

at each inner iteration we solve the linear system

$$\left[I - h_t J(\mathbf{u}_i^{k+1}) \right] \mathbf{p}_i(\mathbf{u}_i^{k+1}) = \mathbf{u}^k + h_t \mathbf{f}(\mathbf{u}_i^{k+1}) - \mathbf{u}_i^{k+1}$$

for $\mathbf{p}_i(\mathbf{u}_i^{k+1})$ and then compute the Newton-Raphson update as:

$$\mathbf{u}_{i+1}^{k+1} = \mathbf{u}_i^{k+1} + \mathbf{p}_i(\mathbf{u}_i^{k+1})$$

What have we learned in this course

- Three classes of 2nd order PDE's: elliptic, parabolic, hyperbolic. Linear, varying coefficients, non-linear. Maximum Principle, uniqueness, positivity.
- Two methods of spatial discretization: FDM and FVM. Local approximation error, global solution error, properties of system matrices.
- Numerical solution of time-dependent problems: Forward- and Backward-Euler (diffusion equation), time-stepping (wave equation). Convergence, stability.
- Numerical methods for nonlinear PDE's: Picard's and Newton-Raphson methods, Banach's Fixed-Point theorem.
- How to program all this in Python.
- **Still to learn:** function spaces, variational calculus, Finite-Element Method, coupled problems, vector- and tensor-valued unknowns, linear solvers, uncertainty analysis, packages, libraries...